

KEVIN COYLE
SOFTWARE ENGINEER | MACHINE LEARNING
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SUMMARY

I am a machine learning engineer, with several years of experience in the full stack of software engineering. I am comfortable with single-node and distributed applications. I have built and maintained end-to-end machine learning and generative AI based software for Fortune 50-100 companies.

PROFESSIONAL EXPERIENCE

AI/ML Technical Lead Engineer

A&M | Los Angeles, CA | 2023 - Present

- Lead application building and deployment for ML projects, along with our research and development efforts in generative AI.
- Lead Senior Data Scientists and Junior Data Scientists with solution architecture and code guidance.
- Notable projects: lead engineer in AWS MySQL to Snowflake (with Airflow pipelines); lead engineer for RAG based personalized recommender for hotel chain, resulting in 30% reduction in call center costs; lead architect and engineer for text2SQL dashboard built in Next.js deployed on Kubernetes with Terraform; lead developer for agriculture client, a Typescript AWS application to track plant genetics and cross-breeding; AWS infrastructure rebuild for eCommerce company, creating \$750k in annual savings; migration internal initiatives including: technical coding environment for recruiting; thought pieces on LLMs.

Adjunct AI/ML Instructor

Caltech | Los Angeles, CA | 2024 - Present

- As an adjunct instructor in the CTME program, I teach a variety of students in topics like Software Engineering for AI, Deploying AI at Scale (with PySpark and related distributed tools), Containerization for AI, and AI on AWS.
- I built and continue to regularly maintain a large portion of the curriculum in courses.

Solution Architect AI/ML

Databricks | San Francisco, CA | 2020 - 2023

- Created reference architectures for machine learning and data engineering pipelines.
- Designed and deployed machine learning models and data pipelines using the Databricks platform with customer specs and requirements.
- Solved customer queries and issues with distributed and single-node machine learning and deep learning.
- Notable projects: end-to-end design of Data Engineering pipeline for a Fortune 50 company. Optimized Apache Spark code to solve issues with skew and spill, saving several customers > \$1.2mn in compute costs.

Other notable, relevant experience:

Senior Data Scientist

Capgemini | Burbank, CA | 2018 - 2020

Data Scientist

Alo Yoga | Commerce, CA | 2017 - 2018

Data Science Lead Instructor

General Assembly | Santa Monica, CA | 2017 - 2022 (part-time)

EDUCATION

California State University, Sonoma, Rohnert Park, CA

Bachelor of Arts, International Relations

CERTIFICATIONS

Databricks ML Associate | Databricks DE Associate | Apache Spark Associate | AWS ML | AWS Solution Architect | Salesforce Data Cloud

SKILLS

Analytics Software Stack:

Python | Scala | R | SQL | NumPy | Pandas | SciKit-Learn | SciPy | PyTorch | MLflow | Databricks | Apache Spark | Delta Lake | Databricks Feature Store | Snowflake | Amazon Redshift | Airflow | Tableau | Looker | Periscope | OpenAI | Langfuse | LangChain

General Tech Stack: Docker | Amazon AWS Suite | Azure suite | Git | Terraform | Kubernetes

Back End / Database Software Stack:

Python | TypeScript | Rust | NodeJS | ExpressJS | Apollo | Graph | PostgreSQL | neo4j | MongoDB | Redis | PineconeDB | PGVector | ChromaDB | FAISS | Selenium | Flask | Django/DRF

Front End Software Stack:

TypeScript | JavaScript | React | NextJS | Husky | Tailwind | HTMX | Vite | Vue/Nuxt | Svelte | ReactNative | Jest | Playwright

Other Skills:

Salesforce Marketing Cloud | Salesforce Data Cloud | Microsoft Office Suite | Google Office Suite | Comfortable in Ubuntu, Mac OS, and Microsoft